

<JobJiffy-A Service providing Platform>

A Minor Project Synopsis Submitted to



**Rajiv Gandhi Proudyogiki Vishwavidyalaya , Bhopal
Towards Partial Fulfillment for the Award of**

**Bachelor of Technology
(Computer Science and Engineering)**

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Jan-May 2025**

1. Introduction

This platform offers a user-friendly interface where customers can browse available professionals, check real-time availability, book services instantly, and make secure payments. Integrated online chat support ensures smooth communication between users and service providers, while customer reviews and ratings help maintain service quality and trust.

2. Objective

- **Seamless Connection** – To provide a digital platform that efficiently connects users with verified professionals for household services such as plumbing, electrical work, carpentry, and cleaning.
- **Real-Time Availability** – To enable users to check and book services based on the real-time availability of professionals, ensuring quick and hassle-free scheduling.
- **Secure Payment System** – To integrate a secure and reliable payment gateway that facilitates smooth transactions while protecting user data.
- **Enhanced Communication** – To offer an online chat feature that allows users and service providers to communicate easily, discuss service requirements, and clarify doubts before booking.
- **Trust and Transparency** – To implement customer reviews and ratings that help users make informed decisions while ensuring service quality and accountability among professionals.

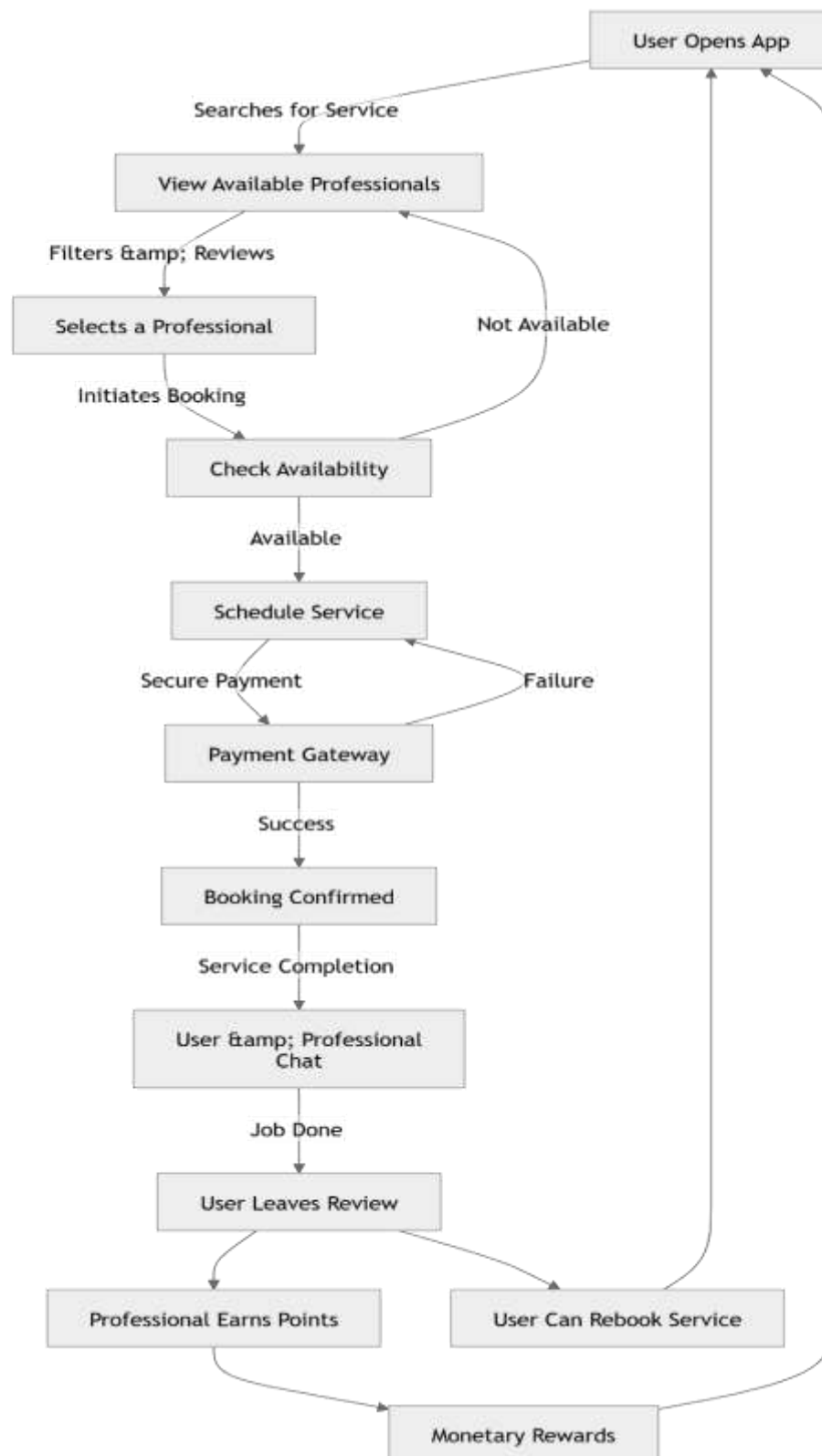
3. Scope

The Freelance Service Booking Platform is designed to streamline the process of hiring skilled professionals for household services by providing a digital marketplace for both users and service providers. The platform's scope covers various functional and non-functional aspects to ensure a seamless and efficient experience.

4. Study of Existing Systems

System	Characteristics	Advantages	Disadvantages
Task Hub	On-Demand service booking for household tasks	Wide service range, user-friendly interface	High service fees, limited regional coverage
HandyConnect	Verified professionals for home maintenance	Strict verification, trusted service providers	Higher pricing, fewer professionals in small towns
QuickFix	Instant booking for urgent household repairs	Fast response time, easy cancellation policy	Limited to emergency services only
ServicePro	AI-driven job matching system	Some job allocation, efficient scheduling	Complex algorithm may misassign professionals
HomeHelp	Subscription-based service platform	Cost-effective for regular users, discounted rates	Requires long term commitment

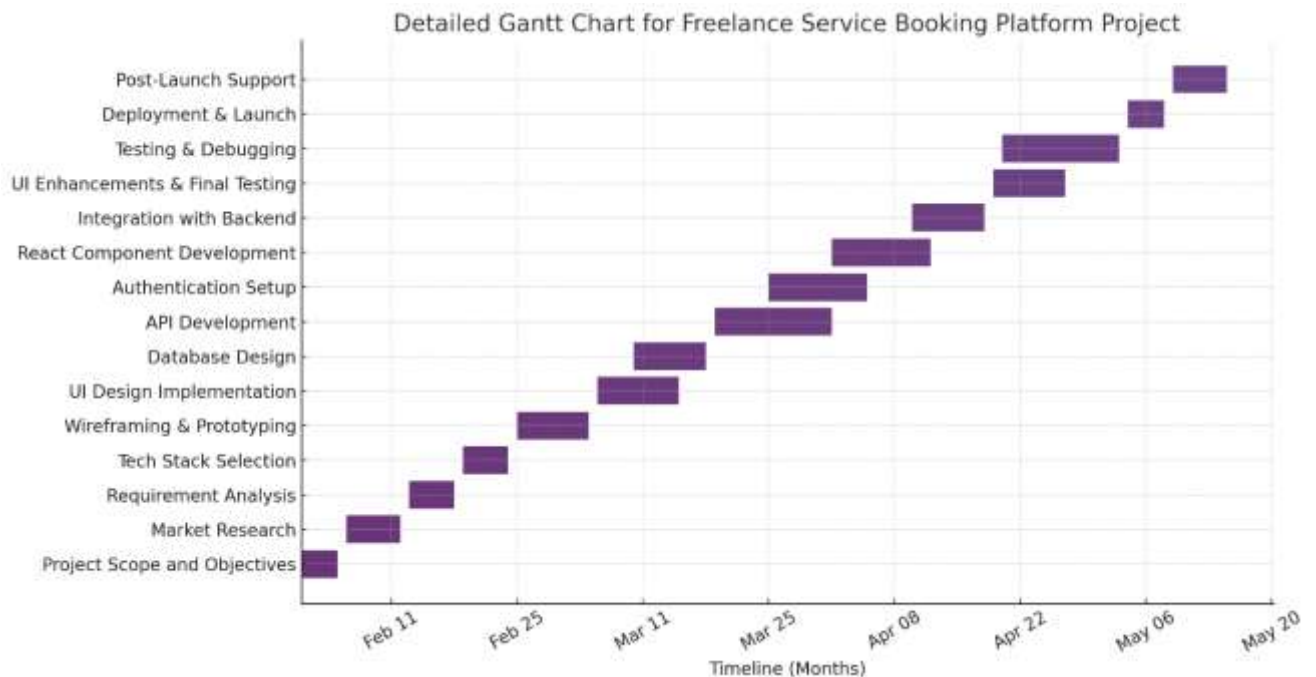
5. Project Description



6. Planning of the Project Work

- **Objective:** Enhance hospital operations by automating processes like patient registration, appointment scheduling, billing, and inventory management.
- **Goal:**
 - Improve patient data management.
 - Optimize hospital resource allocation.
- **Requirements Gathering:**
 - Consult with stakeholders (hospital staff, doctors, IT admins) to identify needs.
 - Define functional requirements (e.g., patient records, appointment scheduling) and non-functional requirements (e.g., data security, scalability).
 - Select technology stack: Java/Python for backend; HTML/CSS/JavaScript for frontend; MySQL/MongoDB for database.
- **System Design:**
 - Create use case and data flow diagrams to visualize system interactions and data movement.
 - Implement a modular architecture for scalability and maintainability.
 - Plan integration with external systems (insurance providers, labs) via APIs.
- **Development Phase:**
 - Follow Agile methodology, organized into 2-3 week sprints focusing on milestones.
- **Testing:**
 - Conduct unit testing, integration testing, and user acceptance testing (UAT) with hospital staff.
- **Deployment:**
 - Deploy on hospital servers or cloud platforms with user training and documentation.
- **Maintenance:**
 - Provide ongoing support and establish feedback mechanisms for continuous improvement.

Gantt chart



7. Features

1. User Management

- User Registration & Authentication – Secure sign-up/login for both customers and service providers using email, phone number, or social media.
- User Profiles – Customers and service providers can create and manage detailed profiles, including past services and reviews.

2. Service Booking & Scheduling

- Service Listings & Categorization – Organized services such as plumbing, electrical work, carpentry, and cleaning.
- Real-Time Availability – Users can check professionals' availability and book services instantly.
- Instant & Scheduled Bookings – Allows users to book services immediately or schedule for a later date.

3. Secure Payment System

- Multiple Payment Options – Supports payments via credit/debit cards, e-wallets, UPI, and net banking.
- Secure Transactions – Uses encrypted payment gateways to ensure safety.
- Transparent Pricing – Displays estimated costs before booking confirmation.

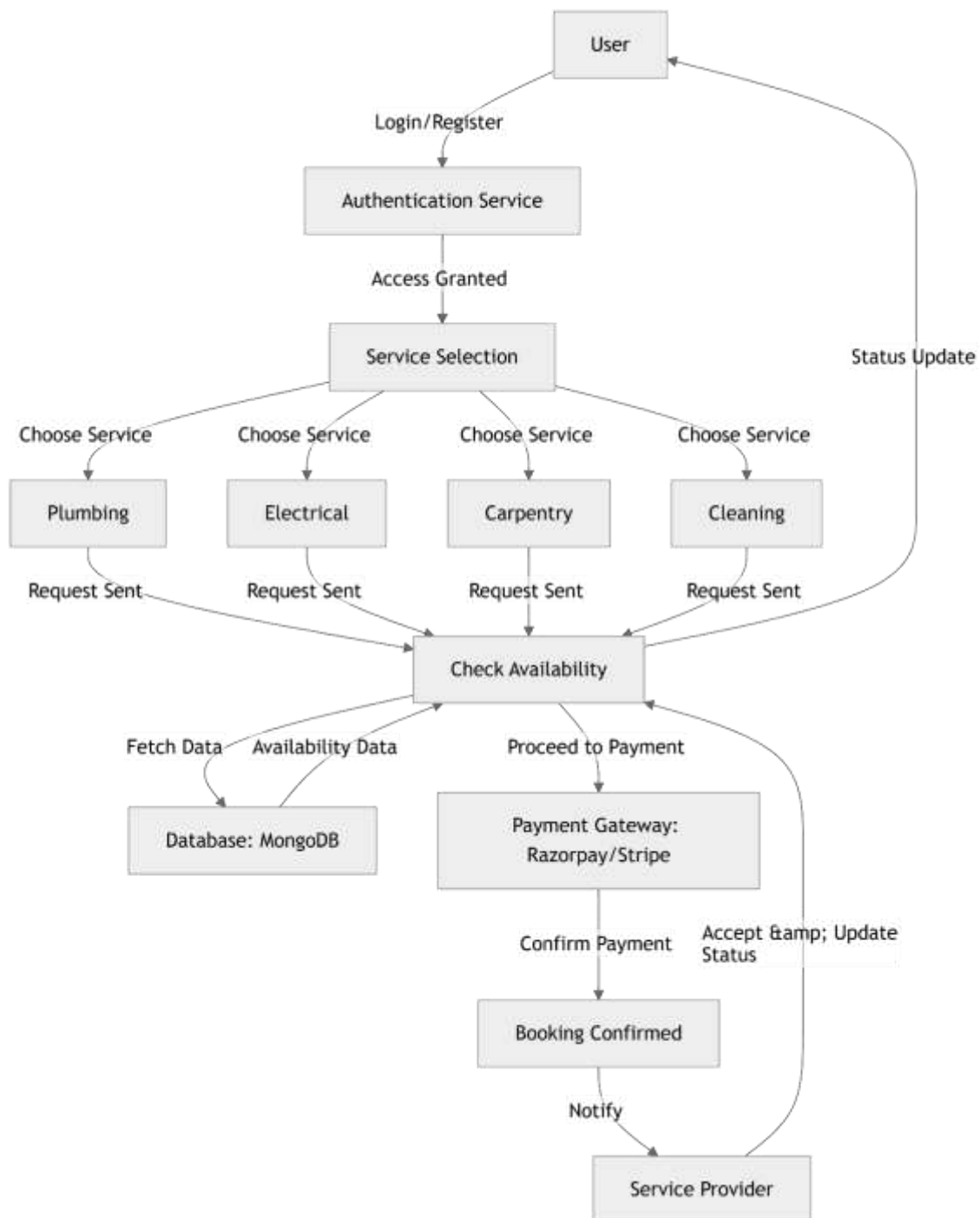
4. Communication & Support

- In-App Chat & Call – Direct messaging and calling features to discuss service requirements.
- Push Notifications & Alerts – Automatic reminders for booking status, payments, and service updates.
- Customer Support & Dispute Resolution – In-app help center for complaints, refunds, and queries.

5. Review & Rating System

- Customer Reviews & Ratings – Users can provide feedback and rate professionals based on service quality.

8. System Architecture



9. Technology Stack

9.1: Programming Languages:

- **JavaScript/React.js:** For client-side scripting and creating interactive elements on the frontend.
- **HTML, CSS:** For frontend development with responsive design and additional styling features.
- **Node.js:** For backend development, handling API requests, data processing, and server-side logic.
- **JavaScript (ES6+):** Used for both frontend and backend scripting to ensure consistency in development.

9.2: Frameworks:

- **Frontend:**
The user interface is developed using **React.js**, which enables a dynamic and user-friendly experience. **HTML and CSS** are used for structuring and styling the application, ensuring responsiveness and accessibility.
- **Backend:**
The backend is powered by **Node.js and Express.js**, facilitating fast and efficient data handling, request processing, and API management.

9.3: Databases:

A MongoDB database is used to store all service provider details, user requests, transactions, and feedback for seamless data retrieval and management.

10. Expected Outcome

1. Enhanced User Experience

- **Easy Booking Process** – Users can quickly find and book services based on real-time availability.
- **Secure Transactions** – Integrated payment gateways ensure safe and hassle-free payments.
- **Transparent Service Ratings** – User reviews and ratings improve service quality and reliability.

2. Increased Job Opportunities for Professionals

- **Flexible Work Schedule** – Service providers can set their availability and accept jobs at their convenience.
- **Improved Income Stability** – Reliable job bookings lead to increased earnings for freelancers.
- **Trust & Credibility** – Verified profiles help professionals build a strong reputation.

3. Efficient Platform Management

- **Automated Booking & Scheduling** – Reduces manual work for both users and service providers.

- **Seamless Communication** – In-app chat and notifications ensure smooth coordination.
- **Data-Driven Insights** – Admin dashboard provides analytics on service trends, user engagement, and revenue.

4. Market Growth & Expansion

- **Scalable Architecture** – The platform can expand to multiple regions and new service categories.
- **AI-Powered Recommendations** – Improves job matching and enhances customer satisfaction.
- **Multi-Language & Localization Support** – Expands reach to a global audience.

5. Higher Customer Satisfaction & Retention

- **Quick & Reliable Services** – Faster response times lead to improved customer trust.
- **Personalized User Experience** – AI-driven suggestions and history-based recommendations.
- **Loyalty Programs & Discounts** – Encourage repeat bookings

13. Resources and Limitations

Technological Resources:

- **Development Tools:** Integrated Development Environments (IDEs) like Visual Studio Code, IntelliJ, or Eclipse for efficient coding and debugging.
- **Frameworks:** Web development frameworks such as React.js, Express.js, and Node.js to streamline UI/UX design and backend functionalities.
- **Database Management Systems:** MongoDB for scalable and efficient data storage and management.
- **Payment Gateway:** Razorpay or Stripe for secure and seamless payment transactions.
- **Testing Tools:** Software like Postman, Selenium, and JUnit for API testing, automation, and unit testing to ensure system reliability.
- **Cloud Services:** AWS or Firebase for hosting, authentication, and cloud-based storage solutions.

Limitations

- Real-time Scheduling Challenges
- Time Constraints
- Payment Facility Integration in Progress
- User Resistance to Adoption

14. Conclusion

In conclusion, The Freelance Service Booking Platform is designed to bridge the gap between users in need of household services and skilled professionals looking for job

opportunities. By leveraging real-time availability, secure payments, instant communication, and customer reviews, the platform ensures a seamless, trustworthy, and efficient experience for both service seekers and providers.

With an AI-driven recommendation system, secure transactions, and scalable cloud infrastructure, the platform has the potential to become a leading solution in the on-demand service industry. However, challenges such as user trust, professional availability, and competition must be carefully addressed through robust security measures, user education, and marketing strategies.

Ultimately, the platform will empower skilled workers by offering flexible employment opportunities while providing users with easy access to reliable home services, creating a win-win ecosystem for all stakeholders

15. References

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